

Efficacy of a Computer Tutor that Models Expert Human Tutors

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Abstract. The contribution of expertise to tutoring effectiveness is unclear and continues to be debated. We conducted a 9-week learning efficacy study of an intelligent tutoring system (ITS) for biology modeled on expert human tutors with two control conditions: human tutors who were experts in the domain but not in tutoring and a no-tutoring condition. All conditions were supplemental to classroom instruction, and students took learning tests immediately before and after tutoring sessions as well as delayed tests 1-2 weeks later. Analysis using logistic mixed-effects modeling indicates significant positive effects on the immediate post-test for the ITS ($d = .71$) and human tutors ($d = .66$) which are in the 99th percentile of meta-analytic effects, as well as significant positive effects on the delayed post-test for the ITS ($d = .36$) and human tutors ($d = .39$). We discuss implications for the role of expertise in tutoring and the design of future studies.

Keywords: intelligent tutoring systems · expert tutor · dialogue · biology

1 Introduction

Meta-analyses of decades of research support the effectiveness of human tutoring for promoting learning [9, 7, 14, 24, 27]. The median effect size (ES) across these meta-analyses is .4 over conventional instruction, which is equivalent to an improvement from the 50th percentile to the 66th percentile. While these meta-analyses don't contrast the effectiveness of tutors by expertise, untrained tutors (.36 ES) are about as effective as trained tutors (.41 ES) [7], and peer tutors (.52 ES) are about as effective as adults (.54 ES) [9]. Indeed across these meta-analyses, the median effect size when children

are tutors (.4 ES) is comparable to the median effect size when adults are tutors (.38 ES). Thus these meta-analyses bring into question the intuition that training and age should increase tutoring effectiveness, leading some to argue that tutor experience is not important as long as the tutor is sufficiently knowledgeable and interactive [27].

Bloom's landmark 2-sigma paper [5] describes tutoring effects that are five times the median tutoring effect above (an improvement from the 50th percentile to the 98th percentile) and has been viewed as evidence supporting the importance of tutoring expertise and promoting "super tutors." Bloom's paper summarizes the work of two of his students [1, 6] who contrasted tutoring, mastery learning (students could not move on until achieving criterion mastery), and conventional instruction. Bloom's 2 ES effect finding, however, obfuscates that his students' studies included a mastery learning criterion in the tutoring condition of 90% vs. 80% in the mastery learning condition [27, 11]. Assuming no interaction between mastery learning and tutoring, we can subtract the mastery effect from the tutoring effect in these studies ($1.91 \text{ ES} - 1.06 \text{ ES} = .85 \text{ ES}$) [1, 6, 10]. While .85 is a larger effect of tutoring than might be expected from meta-analyses generally, it is consistent with a meta-analysis .83 ES reported for human tutoring studies that create their own learning outcome measures when children are tutors [7]. Moreover, the Bloom studies used undergraduate students as tutors, with one study describing tutor training as lasting one week [6]. Altogether, a thorough inspection suggests that the 2-sigma tutors were not "super tutors" after all, yet the 2-sigma paper has been influential in shaping the development of ITSs moving forward, cf. [2].

Intelligent tutoring systems (ITS) compare favorably to human tutors in terms of effectiveness. A meta-analysis of ITS effectiveness found a median .66 ES compared to conventional instruction, which is equivalent to an improvement from the 50th percentile to the 75th percentile [11]. While a median .66 ES for ITS is larger than the median .4 ES for human tutoring described above, the meta-analysis also found that the ES depended heavily on the type of test used to measure learning. When the test was developed by researchers, the ITS effect size was .73 ES, but when a standardized test was used, the ITS effect size was .13 ES. Tests that combined researcher and standardized items had an intermediate .45 ES. Test type was found to be the single most important predictor of ITS effect size, such that when test type was held constant, no other study features influenced the size of the effect. In summary, human tutors may be more effective than ITS when test type is considered, so direct comparisons of ITS and human tutors are essential for understanding their relative effectiveness.

Our research builds on the intuition that expertise contributes to tutoring effectiveness beyond subject matter expertise. This intuition is informed by our observations and analyses of expert human tutors [21, 22]. These expert tutors had 5+ years of tutoring experience, a teaching license, a degree in the tutored subject, and reputations for being effective tutors. In contrast to novice tutors, these expert tutors were more interactive, diagnostic, and gave more discriminating feedback, all of which have been cited as theoretical reasons for the effectiveness of human tutors (for a review see [27]). Our goal was to build an ITS modeled on these expert human tutors, both to increase the effectiveness of the ITS and to provide further evidence on the role of expertise in tutoring. In this paper, we present the results of a 9-week study that compared the ITS to subject matter expert tutors and a classroom control. Our research questions are: (1)

is learning in the ITS condition different from the classroom control and (2) is learning in the ITS condition different from subject matter expert tutors?

2 Guru: an ITS modeled on expert human tutors

We developed an intelligent tutoring system (ITS) for high school biology called Guru. Guru is a dialogue-based ITS in the AutoTutor ITS family [17]. Like AutoTutor, an animated tutor and the student interact using natural language dialogue as well as a multimedia workspace that displays and animates images relevant to the conversation. Student responses are interpreted using natural language understanding to provide formative feedback and tailor the dialogue to individual students' knowledge levels. In contrast to AutoTutor, Guru's pedagogical and motivational strategies are informed by in-depth observation and computational modeling of expert tutoring sessions [23, 21, 8]. The Guru animated pedagogical agent is also more sophisticated than the standard AutoTutor agent and uses motion capture data to produce realistic gestures and pointing. Guru was designed to cover Tennessee Biology I Curriculum Standards using curriculum maps provided by the state. In addition to covering topics in tutorial dialogue sessions, Guru presents interactive tasks such as generating summaries, completing concept maps [15, 16], and cloze tasks [26]. The structure of a Guru tutoring session has been described in previous publications [21, 8, 20, 19].

3 Method

3.1 Participants

Thirty-four tenth graders from an urban high school in the U.S. participated in the study. All students were enrolled in Biology I and had the same teacher for that course. Once a week (for nine weeks) students participated in the study during another class period. Of note, all students were required to pass the state-mandated end-of-course assessment for Biology I to graduate from high school.

3.2 Design and Knowledge Assessments

The study used a three-condition repeated-measures design in which each student interacted with both Guru (ITS) and a human tutor (Human) in addition to their regular classroom instruction (Class). The tutoring topics (for both ITS and Human) always lagged behind what the biology teacher covered in the classroom by one week. In a particular week, students would receive either ITS or Human tutoring (e.g., ITS week 1, Human week 2, or vice versa). The nine-week study included three, 3-week cycles. Each cycle addressed four topics, where two were tutored (A, B) and two were not tutored (X, Y); this enabled comparison of tutoring vs. classroom instruction only. For the first two weeks of a cycle, the tutoring conditions completed immediate pre- and post-tests for both tutored and non-tutored topics (AX or BY). On the third week of a cycle, students took a delayed post-test covering all four topics (ABXY).

The knowledge assessments were multiple-choice tests where items were either obtained from previous state standardized tests across the U.S. or were created by a researcher who was blind to condition. The ratio of researcher-created to standardized items was approximately 2:1. Twelve item pre- and post-tests were administered at the beginning and end of each tutoring session for both the ITS and Human conditions to assess prior knowledge and immediate learning gains, respectively. Half of the items on each test were on a tutored topic and the other half on an untutored topic (Class condition). Test items and order of presentation were randomized across pre- and post-tests across students. On the third week of each cycle, students completed a 48-item delayed post-test. Half of the items on this test were previously seen by students (e.g., on the immediate pre- or post-test) and half the items were new but on the same topics.

Our analysis of the first 3 weeks of the study used repeated measures ANOVA [20]. However, the design and authentic setting of the study were a challenge for ANOVA, particularly since students increasingly missed sessions. Additionally, test items had varying difficulties, and test items were randomly assigned to tests for each participant. Logistic mixed models addressed these concerns by easily handling missing data and modeling the variability of test item correctness clustered by participant (cf. ability) and clustered by item (cf. difficulty). The contrast of interest for the effectiveness of ITS relative to Human and Class conditions is therefore the condition by test interaction using correctness of response on each test item as the dependent variable.

3.3 Procedure

Each tutorial session involved (a) students completing the pretest for 10 minutes (b) tutoring with the ITS or the human tutor for 35 minutes, and (c) the immediate post-test for 10 minutes. The delayed posttest occurred one week after all tutoring was complete. Interactions with each human tutor occurred in groups of two to four, which does not appear to reduce the effectiveness of typical tutors [13]. The tutors were undergraduate biology majors; one was a recent graduate. The tutors were provided with the topic to be tutored, the list of ITS concepts for the topic, and the class biology textbook. Students interacted with the ITS one-on-one using researcher-provided laptops.

Table 1. N per week

Cycle	Week 1	Week 2	Week 3
1	33	31	30
2	25	29	26
3	25	28	26

Note: $N = 34$

Table 2. Assessment descriptive statistics.

Condition	Class		Human		ITS	
	M	SD	M	SD	M	SD
Sessions	2.79	.59	2.68	.73	2.35	.85
Researcher items	.71	.09	.78	.14	.63	.18
Standardized items	.31	.04	.24	.14	.40	.16
Pre-test	.44	.50	.43	.50	.41	.49
Post-test	.49	.50	.63	.48	.64	.48
Delayed test	.40	.49	.56	.50	.55	.50

4 Results

Generally 75-90% of students attended sessions in a given week, as shown in Table 1. Table 2 shows assessment variable means, standard deviations, and item proportions. We fit a logistic mixed model and conducted statistical tests to answer our first two research questions. The model with fixed effects was $correctness \sim condition * test$, where condition was Class, Human, or ITS, and test was Pre-test, Post-test, or Delayed. Maximal random effects that resulted in model convergence [3] yielded an intercepts-only model: $correctness \sim condition * test + (1|participant) + (1|item)$. A Type III ANOVA analysis of the converging model [25] revealed a significant main effect of condition, $\chi^2(2) = 10.83, p = .004$, a significant main effect of test, $\chi^2(2) = 117.13, p < .001$, and a significant interaction between condition and test, $\chi^2(4) = 43.50, p < .001$. To answer our first two research questions, we conducted contrasts within the interaction using Tukey's p-value adjustment [12]. Contrasts of test scores within conditions were conducted to reveal learning effects. The ITS condition had significant differences between pre- and post-test, $z = 7.60, p < .001$, pre- and delayed test, $z = 3.90, p < .001$, and post- and delayed test, $z = -3.88, p < .001$, such that the ITS condition was more likely to correctly answer questions on the post- than the pre-test, $OR = 3.13, d = .71$, and on the delayed test than the pre-test, $OR = 1.77, d = .36$, but less likely to correctly answer questions on the delayed test than the post-test, $OR = .56, d = -.35$. The Human condition likewise had a significant difference between pre- and post-test, $z = 7.71, p < .001$, pre-test and delayed test, $z = 4.34, p < .001$, and post- and delayed test, $z = -3.12, p = .005$, such that the Human condition was more likely to correctly answer questions on the post- than the pre-test, $OR = 2.90, d = .66$, and on the delayed test than the pre-test, $OR = 1.86, d = .39$, but less likely to correctly answer questions on the delayed test than the post-test, $OR = .63, d = -.27$. The Class condition had no significant differences between pre-test and post-test, $z = 2.27, p = .06$, or pre- and delayed test, $z = -1.86, p = .150$. However, there was a significant difference between the delayed test and the post-test, $z = -3.57, p = .001$, such that the Class condition was less likely to correctly answer questions on the delayed test than the post-test, $OR = .62, d = -.30$. Contrasts between conditions across tests similarly followed the pattern of $[ITS = Human] > Class$. There were no significant differences at pretest across conditions, and there were no significant differences at post- or delayed test between ITS and Human conditions. However, at post-test ITS was more likely to answer questions correctly than Class, $z = 4.24, p < .001, OR = 2.24, d = .50$, as was the Human condition, $z = 3.84, p < .001, OR = 2.04, d = .44$. Similarly, at delayed test, ITS was more likely to answer questions correctly than Class, $z = 4.24, p < .001, OR = 2.04, d = .45$, as was the Human condition, $z = 4.42, p < .001, OR = 2.11, d = .47$.

4.1 Discussion

Our research questions in this study were (1) is learning in the ITS condition different from the classroom control and (2) is learning in the ITS condition different from subject matter expert tutors? For the first question, the ITS condition improved learning on both the immediate post- and delayed-test, but the Class showed no significant improvement in learning. These results suggest that the ITS condition is more effective than the

Class condition at promoting learning. For the second question, both ITS and Human conditions had approximately the same effects on learning. The ITS had a slightly larger effect on pre- to post-test ($d = .71$) compared to Human ($d = .66$), and Human had a slightly larger effect on pre- to delayed test ($d = .39$) compared to ITS ($d = .36$), but the ITS and Human conditions were not significantly different at post- or delayed tests. Additionally, the pre- to post-test effect for the ITS that used both standardized and researcher-created test items was in the 99th percentile [11], and the pre- to post-test effect for the human tutors was in the 99th percentile of meta-analytic effects [9, 7, 14, 24, 27] when the 2-sigma studies confounding mastery learning with tutoring were removed. These results suggest that the ITS and Human conditions were both highly and equally and effective at promoting learning.

Our study has several limitations. Since the Class condition was based on a single teacher, a better (or worse) teacher would have affected pre-test scores and corresponding comparisons within and between conditions. Additionally, we were unable to analyze test performance on standardized test items separately because these were both unevenly distributed across topics and randomly assigned across tests, so it is possible we would see a different pattern of results across only standardized items. Finally, the greatest limitation of the study is the lack of contrast between Human and ITS conditions. Our intuition for this research is that expertise in tutoring makes a contribution to tutoring effectiveness beyond subject matter expertise. It was not practical for us to include either a novice ITS matched for content or expert human tutors with the experience criteria described in Section 1 as comparison conditions, but these contrasts would have further informed the role of expertise in tutoring.

One of the challenges in expert tutoring research is the construction of assessments. The expert tutors we have studied base instruction on student needs dynamically [21], but standard research practice is to control for content and use the same assessments across conditions. We argue that controlling for content across conditions effectively forces the use of a curriculum script and would create a handicap for expert human tutor effectiveness. Recent advances in generative AI for assessment suggest that large language models can dynamically produce multiple choice questions and that these questions have comparable quality and psychometric properties to human-authored questions on the same topic [18, 4]. If effective learning assessments can be dynamically generated by AI, then future studies could study both human tutors who were experts in the domain but not in tutoring and expert human tutors in their natural contexts, across topics. Such research would better inform our understanding of the role of expertise in tutoring: not just what expert human tutors do, but how those differences translate into effectiveness.

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